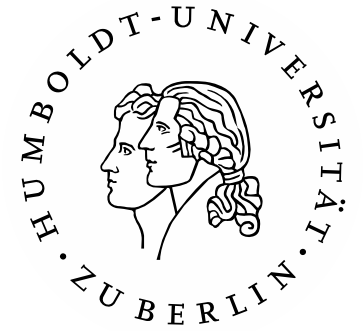




Jacqueline Sachse, Kristina Eichel, Louis Schiekiera, Sophie Müller,
Helen Niemeyer, Felicitas Heßelmann

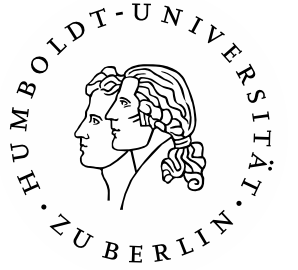
Publication Bias in Academic Publishing: Researchers' Accounts of Publication Decisions

STS-hub 2023, Aachen



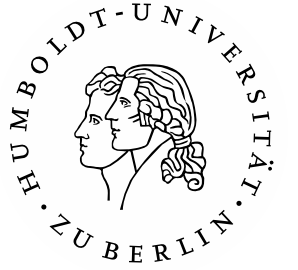
Project PANNE

- “Publication Bias: Analysis of Non-Publication and Non-Reception of Results in Interdisciplinary Comparison”
- funded by Berlin University Alliance (BUA)
- funding period: 2022 – 2023
- cooperation partners:
 - Humboldt-Universität zu Berlin (Robert K. Merton Center for Science Studies)
 - Freie Universität Berlin (Division of Clinical Psychological Intervention)



Publication Bias

- „selective publishing“ of study results (Begg 1994): preference of statistically significant results or results that support existing expectations
- consequences
 - scientific record overrepresents positive results
 - researchers overestimate effect sizes and draw false-positive conclusions
- relation between publishability ↔ research quality ↔ career / reputation



Field-specific publication and reception practices

- publication and research processes are deeply intertwined: working towards publications structures the whole research process (Latour & Woolgar 1987)
- research and publication as social processes
 - collaboration, exchange with colleagues
 - assessment of research quality results from social negotiation processes (Lamont 2009)
- publication bias is a known problem in quantitative disciplines, such as psychology
 - contrast clinical psychology with qualitative and hermeneutic methods as in contemporary history



Research questions

- How does selective (non-)publication emerge?
 - What are social and structural mechanisms in the academic publishing and reward system with regard to publication and reception decisions?
 - Which criteria guide publication and reception decisions where in the research process?



Methods

In progress

- online experiment on publication and reception preferences
- meta review comparing number of positive/negative results within respective discipline

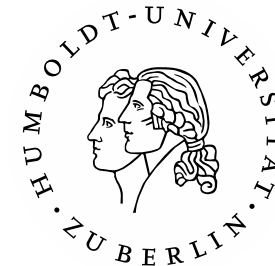
In analysis

- semi-structured one-hour interviews with researchers from clinical psychology and contemporary history
- observation of research and publication processes in a research team (psychology) and a doctoral colloquium (history)



Interviews

- 21 interviews (Jul-Dec 2022)
 - 11 contemporary history, 10 psychology
 - 10 female, 11 male
 - 7 predocs, postdocs and profs
- preliminary analysis of 7 interviews



Different research & publishing cultures

	history	psychology
most important publication format	monograph	journal article
authorship	single author	multiple authors
methods	qualitative, hermeneutical	quantitative, experimental
typical research process	thesis → source analysis → thesis	hypothesis → experiment → confirmation
understanding of “quality”	originality methodological soundness	excellence originality open science

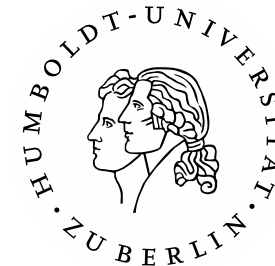


You don't simply work through your thesis. [...] The thesis is always changing. That's the fun part of it. Otherwise, you are not unbiased («ergebnisoffen«).

History professor

When I review, I react very sensitive if I get the feeling that they wrote the hypothesis *after* they have seen the results.

Psychology professor



Different research & publishing cultures

	history	psychology
most important publication format	monograph	journal article
authorship	single author	multiple authors
methods	qualitative, hermeneutical	quantitative, experimental
typical research process	thesis → source analysis → thesis	hypothesis → experiment → confirmation
understanding of “quality”	originality methodological soundness	excellence originality open science



Discipline-specific mechanisms: history (1)

- access to publication venues difficult without personal network
 - book chapters and articles are mostly invited
 - doctoral students depend on their supervisors
- no standardized review processes
 - editors strongly shape manuscript revision with own opinions



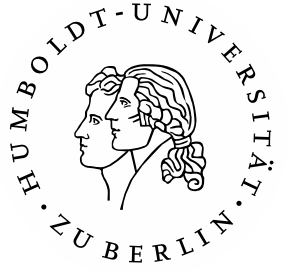
Discipline-specific mechanisms: history (2)

- difficult or failed publication projects:
 - politically sensitive topics or unpopular analytical perspectives
- I06's supervisors discouraged him from studying a certain ethnic group not from the victim's, but perpetrator perspective, and warned him about negative career implications. "Are you sure you want your name connected to these people?"
- The publishing contract of I07's book was cancelled after harsh and emotional criticism at an international conference. "Some of them [other scholars] became very emotional and insulted me personally. After that, [the publisher] said they won't publish my book anymore."



Discipline-specific mechanisms: psychology (1)

- strategic publication planning at the beginning of a project
 - distribution of first-authorships to team members
- careful journal selection: balancing high JIF against chances to be accepted
 - open-science expectations incompatible with high-impact journal requirements
 - “I think it is a bit of a hubris, that it is being demanded that you publish completely and show everything, but then after all it is not wanted and your paper gets rejected because of it.” (postdoc)
 - “I am willing to compromise in terms of reputation or impact factor if [...] the journal pushes you to deposit your data online or do a preregistration. [...] Even if that implied negative consequences for my career.” (doctoral student)



Discipline-specific mechanisms: psychology (2)

- negative findings: hypotheses cannot be confirmed
 - doubt correctness of data, search for methodological weaknesses
 - worry about reviewers or community feeling attacked by results
- explorative analyses: no clear hypotheses
 - worry about “how to sell” research approach and results, and how to construct a convincing narrative
 - emphasize that results are “especially interesting”
 - placing of unfavored results in less prestigious venues



Conclusions: How does selective (non-)publication emerge?

- publication decisions are made within social processes, and publication processes are embedded in established power saturated research practices
 - need to meet field expectations
 - dependence of doctoral students from supervisors
- publication bias is a result of social and hierarchical structures of the academic publishing and reward system, rather than a form of misbehavior of individual researchers



References

Begg, C.B. and Mazumdar, M. (1994) Operating Characteristics of a Rank Correlation Test for Publication Bias. *Biometrics*, 50, 1088-1101.

Lamont M. (2009) *How Professors Think: Inside the Curious World of Academic Judgment*. Cambridge, Mass: Harvard University Press.

Latour B. and Woolgar S. (1986) *Laboratory Life: The Construction of Scientific Facts*. 2nd Edition with a New Postscript. London: Sage Publications.



Thank you for your attention!

Jacqueline Sachse

Robert K. Merton Center for Science Studies
Humboldt-Universität zu Berlin, Germany
jacqueline.sachse@hu-berlin.de

 <https://orcid.org/0000-0003-2587-4305>